

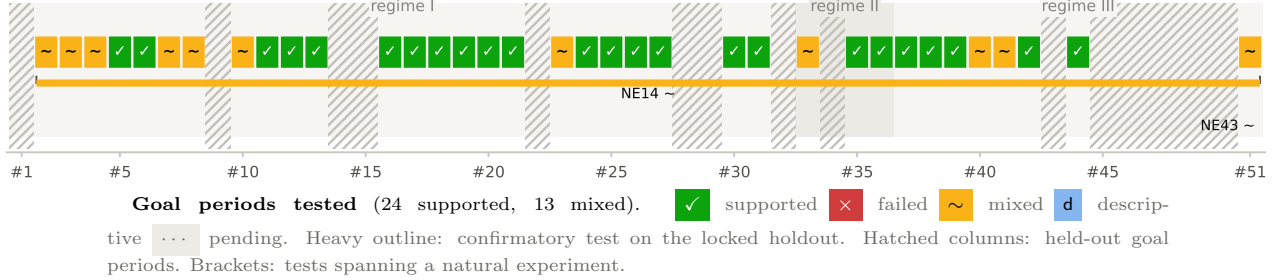
# H50: Field or coupling: transfer functions and lag tell which

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 2.5/5

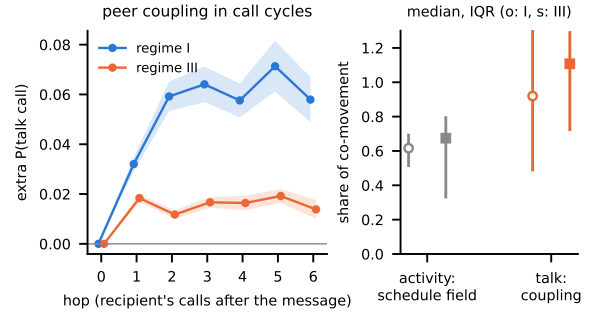
**Bottom line.** In LLM swarms, “moving together” in activity is a field set by the scheduler, while talk propagates by a coupling gated at exactly one read-out call, which agents rarely relay beyond that.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

Do agents co-move because a common input hits everyone (a field) or because they read each other (a coupling)? In call cycles, a field moves every agent at its next call; a coupling acts only at the recipient’s first call that can read the message. We used the DQ1 context ledger to compare calls that start just after a message with calls just before it (a placebo-corrected discontinuity, hop by hop), fitted input impulse responses, and decomposed co-movement on 71 non-holdout units (35 periods), with synthetic swarms first. Result: **activity co-movement is the schedule** (day start and stop explain about 0.6 beyond a null). **Talk co-movement is a gated coupling**: a peer message raises the chance that the next call is a talk call (45/71 units, none negative), always at hop 1, and this accounts for roughly all talk co-movement. HH167’s regime split fails. Holdout not run.



**Observables.** Left: extra probability that a recipient’s call is a talk call, by hop after a peer message (IVW pooled over units; band 95%). Zero at the in-flight call (hop 0), a step at the read-out call (hop 1). Regime III (13-s chained calls) drops at hop 2, then plateaus; regime I (scheduled chat calls) keeps rising. Right: activity co-movement explained by the schedule edges (gray) vs talk co-movement explained by the gated coupling (orange; values > 1 are overshoot).

## PHYSICAL MODEL

Gated linear response (models 02, 01). Each agent makes calls  $c$  at times  $t_c$ ; a message posted at  $t_m$  enters context at the first call with  $t_c > t_m$  (hop 1). Call outcome  $Y_c$  (talk, work, idle) responds to inputs  $u_{c-k}$  (bookends, human messages, nudges, platform errors) and peer messages  $m_{c-k}$  first read  $k$  hops earlier.  $s$ : call start minus message time (bandwidth 1.5 median call intervals);  $e_{it}$ : agent-day-demeaned minute activity or talk;  $S$ : equal-time co-movement.

$$\text{logit } P(Y_c=1) = h_i + \sum_k g(k) u_{c-k} + \sum_k J(k) m_{c-k}$$

$$J_k = E[Y | s \rightarrow 0^+] - E[Y | s \rightarrow 0^-] - \text{placebo}, \quad \delta(h) = \sum_{k \leq h} J_k$$

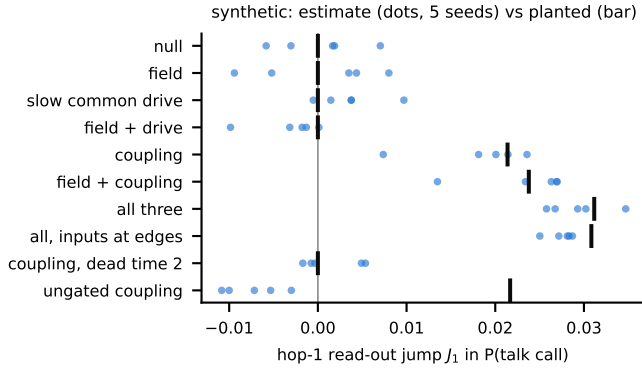
$$S = \sum_t [(\sum_i e_{it})^2 - \sum_i e_{it}^2] / \sum_t \sum_i e_{it}^2$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful, mostly as a correction to how swarm synchrony is read. Most of the collective mode in *activity* is the platform starting and stopping everyone; it says nothing about agents influencing each other. Influence is real but narrow. It acts at the next call, turns a work call into a talk call, and in the always-on regime it mostly needs the recipient to be named (jump 0.17 vs 0.004). For an operator: synchrony in who is busy is a scheduler artifact; to propagate something, address agents by name and expect a response within one call (about 15–25 s, minutes if paused). The nudger moves single agents and leaves the collective mode alone. Bookend messages are announcements, not inputs.

## RESULTS VS PREDICTIONS

| Prediction (dated)           | Observed                                                                                                  | Verdict  |
|------------------------------|-----------------------------------------------------------------------------------------------------------|----------|
| P1 synthetic recovery        | $J_1$ bias $-2$ to $-17\%$ (III); dead time 2 $\rightarrow$ onset hop 3; nulls 3/35                       | passed   |
| P2 inputs at read-out        | in-room human $J_1 > 0$ , hop 1; nudged agents talk at hop 1 (0.14), not later                            | mixed    |
| P3 low-pass, corner 5–40 min | coherence $8-10\times$ at 15–60 min; mixed corner ill-defined (undershoot)                                |          |
| P4 field: zero relative lag  | onsets IQR 9–22 s ( $< 2$ cycles); supported #38 agents halted at pause                                   |          |
| P5 gated coupling, hop 1     | $J_1 > 0$ in 45/71, $< 0$ in 0; onset hop 1; no decay in regime I                                         | mostly   |
| P6 decomposition             | activity field 0.55 (III) $\checkmark$ , 0.51 (I, pred. $\leq 0.2$ ) $\times$ ; talk coupling $> 0.57/68$ | mostly   |
| HH167 regime split           | regime-I calls not message-triggered (ratio 1.03–1.07)                                                    | rejected |
| NE43 switch-offs             | nudge path gone; coupling $\times 0.69$ ; edge field persists                                             | mixed    |
| G51 human relay              | plain human messages: hop-1 field; relay underpowered                                                     | mixed    |
| G38 pause gates              | 82% start before any peer talks; supported 94% never call after pause                                     |          |



**Second figure.** Synthetic swarms at village sampling with real input schedules (5 seeds each). The read-out jump finds planted coupling (with or without fields, slow common drive or edge-clustered inputs), reads zero for fields and drives, recovers a 2-hop dead time as a later onset, and goes negative for ungated coupling.

## GOAL PERIODS IN DETAIL

- G02  $\sim$  mixed
- G03  $\sim$  mixed
- G04  $\sim$  mixed
- G05  $\checkmark$  supported
- G06  $\checkmark$  supported
- G07  $\sim$  mixed
- G08  $\sim$  mixed
- G10  $\sim$  mixed
- G11  $\checkmark$  supported
- G12  $\checkmark$  supported
- G13  $\checkmark$  supported
- G16  $\checkmark$  supported
- G17  $\checkmark$  supported
- G18  $\checkmark$  supported
- G19  $\checkmark$  supported
- G20  $\checkmark$  supported

+21 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 2 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

Regime-I call starts are latency-placed; the jump survives on logged-start recipients (#24–#31; 0.030 vs 0.034), but early regime I is unchecked and its rising kernel beyond hop 2 is unvalidated. Per-unit CIs are mildly anti-conservative (synthetic false positives about 9%). Decomposition shares are model-based and do not add up (coupling amplifies fields);  $f_C > 1$  is overshoot at low talk co-movement. NE43 is confounded by room changes and roster joins. The #51 target and relay tests have about 100 pairs. No content channel yet. Holdout not run.

## ROUND 2

Content gate with matched-age control; validate the regime-I kernel; per-agent “impedance” vs H29/H32 influence; confirm on the holdout, including NE23 (nudger off/on).

- **Essence.** which levers move a swarm together (fields) and which only propagate through who reads whom (couplings), measured in the units the scaffold imposes (calls), so an operator knows what an input will do to collective behavior.
- **H50-R1** Content gate (C3).
- **H50-R2** Validate the regime-I dead-time kernel.
- **H50-R3** HH188 frequency-dependent T.eff.
- **H50-R4** HH190 agent impedance.
- **H50-R5** Human-input relay with power.
- **H50-R6** HH187 graph clause.
- **H50-R7** Confirm on the holdout (confirm.py: CP1–CP5, NE23 nudger off/on).